

Code Documentation Report

Memory

2026-04-30

Chapter 1 — Code Structure Overview

This codebase is organized into multiple modules that collaborate to parse source files, build a graph, enrich metadata, and generate documentation.

Module Name	Number of Functions	Number of Classes	Category
app/memory/orchestration/interaction_filter.py	8	1	core logic
app/memory/routing/intent_classifier.py	19	1	core logic
app/storage/sqlite_store.py	27	1	core logic
app/api/graph.py	3	0	graph
app/memory/stores/graph_memory.py	0	0	graph
app/storage/graph_store.py	6	1	graph
app/api/chat.py	2	2	orchestration
app/api/debug.py	4	0	orchestration
app/api/router_api.py	3	2	orchestration
app/api/tree.py	2	0	orchestration
app/core/config.py	1	2	orchestration
app/core/embeddings.py	2	1	orchestration
app/core/llm.py	3	1	orchestration
app/core/providers/gemini_provider.py	3	1	orchestration
app/core/providers/openai_provider.py	3	1	orchestration
app/jobs/sleep_consolidation_job.py	4	1	orchestration
app/jobs/start_background_jobs.py	2	0	orchestration
app/main.py	3	0	orchestration
app/mcp/server.py	3	0	orchestration
app/memory/behavior/emotional_analyzer.py	2	0	orchestration
app/memory/consolidation/compaction_engine.py	4	1	orchestration
app/memory/consolidation/memory_lifecycle.py	3	1	orchestration
app/memory/consolidation/reflect_engine.py	2	1	orchestration
app/memory/consolidation/sleep_cycle.py	1	1	orchestration
app/memory/extraction/contradiction_detector.py	5	1	orchestration
app/memory/extraction/memory_extractor.py	7	1	orchestration
app/memory/extraction/memory_validator.py	5	1	orchestration
app/memory/orchestration/attention_engine.py	3	1	orchestration
app/memory/orchestration/decision_engine.py	1	1	orchestration
app/memory/orchestration/episode_engine.py	1	1	orchestration
app/memory/routing/bert_router.py	5	0	orchestration
app/memory/routing/fallback_manager.py	2	1	orchestration
app/memory/routing/oos_detector.py	2	1	orchestration

Module Name	Number of Functions	Number of Classes	Category
app/memory/routing/query_router.py	6	1	orchestration
app/memory/routing/routing_policies.py	1	1	orchestration
app/memory/stores/episodic_memory.py	0	0	orchestration
app/memory/stores/semantic_memory.py	0	0	orchestration
app/memory/stores/working_memory.py	0	0	orchestration
app/memory/supermemory.py	4	1	orchestration
app/schemas/memory_models.py	1	15	orchestration
app/storage/faiss_vector_store.py	7	1	orchestration
app/storage/vector_store.py	5	1	orchestration
app/workflow/events.py	0	1	orchestration
app/workflow/orchestrator.py	5	1	orchestration
app/workflow/stages.py	0	3	orchestration
app/memory/retrieval/context_manager.py	2	1	retrieval/indexing
app/memory/retrieval/context_tree.py	5	2	retrieval/indexing
app/memory/retrieval/hierarchical_summarizer.py	1	1	retrieval/indexing
app/memory/retrieval/hybrid_retriever.py	3	1	retrieval/indexing
app/memory/retrieval/persistent_tree_store.py	5	1	retrieval/indexing
app/memory/retrieval/query_resolver.py	1	1	retrieval/indexing
app/memory/retrieval/retrieval_ranker.py	3	1	retrieval/indexing
app/memory/retrieval/tree_indexer.py	4	1	retrieval/indexing
app/memory/retrieval/tree_retriever.py	4	1	retrieval/indexing

Dependency summary: 1288 graph edges identified.

Chapter 2 — Function Reference

startup_event

Function `startup\_event` executes domain-specific processing. Coordinates downstream calls such as `RuntimeError`, `app.on_event`, `faiss_store.initialize`. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `RuntimeError`, `app.on_event`, `faiss_store.initialize`.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	<code>sqlite_store.initialize</code> , <code>RuntimeError</code> , <code>faiss_store.initialize</code> , <code>start_background_jobs</code> , <code>app.on_event</code>
Route	<code>Memory/app/main.py::startup_event</code>
Tags	Workflow
Logic	Coordinates downstream calls such as <code>RuntimeError</code> , <code>app.on_event</code> , <code>faiss_store.initialize</code> .
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

root

Function `root` executes domain-specific processing. Performs `root` behavior for this module. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	-
Route	<code>Memory/app/main.py::root</code>
Tags	Workflow
Logic	Performs `root` behavior for this module.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

chat

Function `chat` executes domain-specific processing. Implements API/service handler behavior and returns normalized response data. Inputs: payload. Outputs: unspecified return value. Called by: no internal callers detected. Calls: orchestrator.run.

Inputs	payload
Outputs	unspecified return value
Called by	-
Calls	orchestrator.run
Route	Memory/app/api/chat.py::chat
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

recent

Function `recent` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: sqlite_store.list_recent.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	sqlite_store.list_recent
Route	Memory/app/api/debug.py::recent
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

stats

Function `stats` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: no explicit parameters. Outputs:

unspecified return value. Called by: no internal callers detected. Calls: `sqlite_store.interaction_stats`, `sqlite_store.is_available`, `sqlite_store.list_recent`.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	<code>sqlite_store.list_recent</code> , <code>sqlite_store.interaction_stats</code> , <code>sqlite_store.is_available</code>
Route	<code>Memory/app/api/debug.py::stats</code>
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

interactions

Function ``interactions`` executes domain-specific processing. Coordinates downstream calls such as `sqlite_store.list_interactions`. Inputs: `limit`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `sqlite_store.list_interactions`.

Inputs	<code>limit</code>
Outputs	unspecified return value
Called by	-
Calls	<code>sqlite_store.list_interactions</code>
Route	<code>Memory/app/api/debug.py::interactions</code>
Tags	Workflow
Logic	Coordinates downstream calls such as <code>sqlite_store.list_interactions</code> .
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

graph_data

Function ``graph_data`` executes domain-specific processing. JSON graph data: center node + neighbors + labeled edges. Inputs: `memory_id`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `graph_store.rich_neighborhood`.

Inputs	memory_id
Outputs	unspecified return value
Called by	-
Calls	graph_store.rich_neighborhood
Route	Memory/app/api/graph.py::graph_data
Tags	Tagging, Workflow
Logic	JSON graph data: center node + neighbors + labeled edges.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

graph_view

Function `graph\_view` serves application-facing request handling. Interactive visual graph viewer using vis-network. Inputs: memory_id. Outputs: unspecified return value. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	memory_id
Outputs	unspecified return value
Called by	-
Calls	-
Route	Memory/app/api/graph.py::graph_view
Tags	Tagging, Workflow
Logic	Interactive visual graph viewer using vis-network.
In Workflow	Yes
Workflow Reason	Reachable from entry points as part of execution path.

debug_route

Function `debug\_route` serves application-facing request handling. Test the intent classifier and router logic without running the full pipeline. Inputs: req. Outputs: unspecified return value. Called by: no internal callers detected. Calls: query_router.route.

Inputs	req
Outputs	unspecified return value
Called by	-

Calls	query_router.route
Route	Memory/app/api/router_api.py::debug_route
Tags	Workflow
Logic	Test the intent classifier and router logic without running the full pipeline.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

router_stats

Function `router\_stats` serves application-facing request handling. Returns general stats or config of the router. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	query_router.policies.POLICIES.keys
Route	Memory/app/api/router_api.py::router_stats
Tags	Workflow
Logic	Returns general stats or config of the router.
In Workflow	Yes
Workflow Reason	Reachable from entry points as part of execution path.

get_tree

Function `get\_tree` loads required project data. Retrieves module data required by downstream functions. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: persistent_tree_store.load, tree.to_dict.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	persistent_tree_store.load, tree.to_dict
Route	Memory/app/api/tree.py::get_tree
Tags	Workflow

Logic	Retrieves module data required by downstream functions.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

embed

Function `embed` executes domain-specific processing. Coordinates downstream calls such as `math.sqrt`, `ord`. Inputs: `self`, `text`, `dim`. Outputs: `List`. Called by: no internal callers detected. Calls: `math.sqrt`, `ord`.

Inputs	<code>self</code> , <code>text</code> , <code>dim</code>
Outputs	<code>List</code>
Called by	-
Calls	<code>ord</code> , <code>math.sqrt</code>
Route	<code>Memory/app/core/embeddings.py::embed</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>math.sqrt</code> , <code>ord</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as `GeminiProvider`, `OpenAIProvider`. Inputs: `self`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `GeminiProvider`, `OpenAIProvider`.

Inputs	<code>self</code>
Outputs	unspecified return value
Called by	-
Calls	<code>settings.llm_provider.lower.strip</code> , <code>settings.llm_provider.lower</code> , <code>GeminiProvider</code> , <code>OpenAIProvider</code>
Route	<code>Memory/app/core/llm.py::init</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>GeminiProvider</code> , <code>OpenAIProvider</code> .

In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

chat

Function `chat` executes domain-specific processing. Implements API/service handler behavior and returns normalized response data. Inputs: self, system, user, temperature. Outputs: str. Called by: no internal callers detected. Calls: self.backend.chat.

Inputs	self, system, user, temperature
Outputs	str
Called by	-
Calls	self.backend.chat
Route	Memory/app/core/llm.py::chat
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as CompactionEngine, MemoryLifecycle, ReflectEngine. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: CompactionEngine, MemoryLifecycle, ReflectEngine.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	CompactionEngine, ReflectEngine, MemoryLifecycle, TreeIndexer
Route	Memory/app/jobs/sleep_consolidation_job.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as CompactionEngine, MemoryLifecycle, ReflectEngine.
In Workflow	No

Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.
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run_once

Function `run\_once` orchestrates execution flow. orchestrates workflow execution steps
 Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: defaultdict, graph_store.prune, memory_service.store.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	sqlite_store.list_recent, defaultdict, type_groups.append, type_groups.items, self.compactor.compact, memory_service.store, self.reflector.reflect, self.lifecycle.decay, sqlite_store.update_status, self.tree_indexer.build_from_memories, persistent_tree_store.save, sqlite_store.prune_interactions, graph_store.prune
Route	Memory/app/jobs/sleep_consolidation_job.py::run_once
Tags	Workflow
Logic	orchestrates workflow execution steps
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

run_forever

Function `run\_forever` orchestrates execution flow. orchestrates workflow execution steps
 Inputs: self, interval_seconds. Outputs: unspecified return value. Called by: no internal callers detected. Calls: self.run_once, time.sleep.

Inputs	self, interval_seconds
Outputs	unspecified return value
Called by	-
Calls	self.run_once, time.sleep
Route	Memory/app/jobs/sleep_consolidation_job.py::run_forever
Tags	Workflow
Logic	orchestrates workflow execution steps
In Workflow	Yes

Workflow Reason	Reachable from entry points and dispatches work to downstream functions.
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start_background_jobs

Function `start\_background\_jobs` executes domain-specific processing. Coordinates downstream calls such as `_thread.is_alive`, `_thread.start`, `threading.Thread`. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `_thread.is_alive`, `_thread.start`, `threading.Thread`.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	<code>_thread.is_alive</code> , <code>threading.Thread</code> , <code>_thread.start</code>
Route	Memory/app/jobs/start_background_jobs.py::start_background_jobs
Tags	Utilities
Logic	Coordinates downstream calls such as <code>_thread.is_alive</code> , <code>_thread.start</code> , <code>threading.Thread</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

mcp_search

Function `mcp\_search` serves application-facing request handling. Processes `mcp search` request flow and returns structured response data. Inputs: query. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `memory_service.retrieve`.

Inputs	query
Outputs	unspecified return value
Called by	-
Calls	<code>memory_service.retrieve</code>
Route	Memory/app/mcp/server.py::mcp_search
Tags	Workflow
Logic	Processes `mcp search` request flow and returns structured response data.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

mcp_recent

Function `mcp_recent`` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: no explicit parameters. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `sqlite_store.list_recent`.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	<code>sqlite_store.list_recent</code>
Route	<code>Memory/app/mcp/server.py::mcp_recent</code>
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

`_passes_admission`

Function `_passes_admission`` executes domain-specific processing. Gate: only high-value memories enter long-term storage. Inputs: `self, m`. Outputs: `bool`. Called by: no internal callers detected. Calls: `getattr`.

Inputs	<code>self, m</code>
Outputs	<code>bool</code>
Called by	-
Calls	<code>m.content.strip, getattr</code>
Route	<code>Memory/app/memory/supermemory.py::passes_admission</code>
Tags	Utilities
Logic	Gate: only high-value memories enter long-term storage.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

store

Function `store`` executes domain-specific processing. Coordinates downstream calls such as `faiss_store.add, getattr, self._passes_admission`. Inputs: `self, memories`. Outputs: `List`. Called by: no internal callers detected. Calls: `faiss_store.add, getattr, self._passes_admission`.

Inputs	<code>self, memories</code>
Outputs	<code>List</code>

Called by	-
Calls	self._passes_admission, sqlite_store.add_memory, faiss_store.add, time.sleep, sqlite_store.delete_memory, ids.append, getattr
Route	Memory/app/memory/supermemory.py::store
Tags	Utilities
Logic	Coordinates downstream calls such as faiss_store.add, getattr, self._passes_admission.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

retrieve

Function `retrieve` executes domain-specific processing. loads and retrieves required data

Inputs: self, query, limit. Outputs: List. Called by: no internal callers detected. Calls: sqlite_store.search.

Inputs	self, query, limit
Outputs	List
Called by	-
Calls	sqlite_store.search
Route	Memory/app/memory/supermemory.py::retrieve
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as Path, faiss.IndexFlatL2, threading.RLock. Inputs: self, dim, index_path. Outputs: unspecified return value. Called by: no internal callers detected. Calls: Path, faiss.IndexFlatL2, threading.RLock.

Inputs	self, dim, index_path
Outputs	unspecified return value

Called by	-
Calls	Path, threading.RLock, faiss.IndexFlatL2
Route	Memory/app/storage/faiss_vector_store.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as Path, faiss.IndexFlatL2, threading.RLock.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

initialize

Function `initialize` executes domain-specific processing. Coordinates downstream calls such as self._load. Inputs: self. Outputs: no explicit return value. Called by: no internal callers detected. Calls: self._load.

Inputs	self
Outputs	None
Called by	-
Calls	self._load
Route	Memory/app/storage/faiss_vector_store.py::initialize
Tags	Utilities
Logic	Coordinates downstream calls such as self._load.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_load

Function `\_load` loads required project data. loads and retrieves required data Inputs: self. Outputs: no explicit return value. Called by: no internal callers detected. Calls: faiss.IndexFlatL2, faiss.read_index, open.

Inputs	self
Outputs	None
Called by	-
Calls	faiss.IndexFlatL2, self.index_path.exists, faiss.read_index, self.meta_path.exists, open, pickle.load

Route	Memory/app/storage/faiss_vector_store.py::load
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`_persist`

Function `_persist` executes domain-specific processing. Coordinates downstream calls such as `faiss.write_index`, `open`, `pickle.dump`. Inputs: `self`. Outputs: no explicit return value. Called by: no internal callers detected. Calls: `faiss.write_index`, `open`, `pickle.dump`.

Inputs	<code>self</code>
Outputs	None
Called by	-
Calls	<code>self.index_path.with_suffix</code> , <code>self.meta_path.with_suffix</code> , <code>faiss.write_index</code> , <code>open</code> , <code>pickle.dump</code> , <code>tmp_index.replace</code> , <code>tmp_meta.replace</code>
Route	Memory/app/storage/faiss_vector_store.py::persist
Tags	Utilities
Logic	Coordinates downstream calls such as <code>faiss.write_index</code> , <code>open</code> , <code>pickle.dump</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`add`

Function `add` executes domain-specific processing. Coordinates downstream calls such as `embedder.embed`, `np.array`, `self._load`. Inputs: `self`, `item_id`, `text`. Outputs: no explicit return value. Called by: no internal callers detected. Calls: `embedder.embed`, `np.array`, `self._load`.

Inputs	<code>self</code> , <code>item_id</code> , <code>text</code>
Outputs	None
Called by	-
Calls	<code>self.initialize</code> , <code>np.array</code> , <code>embedder.embed</code> , <code>self.index.add</code> , <code>self._persist</code> , <code>self._load</code>
Route	Memory/app/storage/faiss_vector_store.py::add

Tags	Utilities
Logic	Coordinates downstream calls such as embedder.embed, np.array, self._load.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

search

Function `search` serves application-facing request handling. Processes `search` request flow and returns structured response data. Inputs: self, query, top_k. Outputs: List. Called by: no internal callers detected. Calls: embedder.embed, np.array, self.index.search.

Inputs	self, query, top_k
Outputs	List
Called by	-
Calls	self.initialize, np.array, embedder.embed, self.index.search, results.append
Route	Memory/app/storage/faiss_vector_store.py::search
Tags	Retrieval, Workflow
Logic	Processes `search` request flow and returns structured response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

add_relation

Function `add\_relation` executes domain-specific processing. Add an edge, skipping duplicates via unique index constraints. Inputs: self, from_id, to_id, relation, weight. Outputs: no explicit return value. Called by: no internal callers detected. Calls: sqlite_store.add_edge.

Inputs	self, from_id, to_id, relation, weight
Outputs	None
Called by	-
Calls	sqlite_store.add_edge
Route	Memory/app/storage/graph_store.py::add_relation
Tags	Tagging
Logic	Add an edge, skipping duplicates via unique index constraints.

In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

neighborhood

Function `neighborhood` executes domain-specific processing. Get all edges touching this memory node. Inputs: self, memory_id. Outputs: List. Called by: no internal callers detected. Calls: sqlite_store.get_edges.

Inputs	self, memory_id
Outputs	List
Called by	-
Calls	sqlite_store.get_edges
Route	Memory/app/storage/graph_store.py::neighborhood
Tags	Tagging
Logic	Get all edges touching this memory node.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

rich_neighborhood

Function `rich\_neighborhood` executes domain-specific processing. Full graph view: center node + connected nodes + labeled edges. Inputs: self, memory_id. Outputs: Dict. Called by: no internal callers detected. Calls: node_ids.add, self.neighborhood, sqlite_store.get_memory.

Inputs	self, memory_id
Outputs	Dict
Called by	-
Calls	sqlite_store.get_memory, self.neighborhood, node_ids.add, nodes.append, vis_edges.append
Route	Memory/app/storage/graph_store.py::rich_neighborhood
Tags	Tagging
Logic	Full graph view: center node + connected nodes + labeled edges.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

connect_related_memories

Function `connect\_related\_memories` executes domain-specific processing. Connect a set of memories that were extracted from the same interaction. Inputs: self, memory_ids, relation. Outputs: no explicit return value. Called by: no internal callers detected. Calls: self.add_relation.

Inputs	self, memory_ids, relation
Outputs	None
Called by	-
Calls	self.add_relation
Route	Memory/app/storage/graph_store.py::connect_related_memories
Tags	Tagging
Logic	Connect a set of memories that were extracted from the same interaction.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

prune

Function `prune` orchestrates execution flow. orchestrates workflow execution steps Inputs: self, retention_days, max_edges. Outputs: int. Called by: no internal callers detected. Calls: sqlite_store.prune_edges.

Inputs	self, retention_days, max_edges
Outputs	int
Called by	-
Calls	sqlite_store.prune_edges
Route	Memory/app/storage/graph_store.py::prune
Tags	Tagging, Workflow
Logic	orchestrates workflow execution steps
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as threading.RLock, threading.local. Inputs: self, db_path. Outputs: unspecified return value. Called by: no internal callers detected. Calls: threading.RLock, threading.local.

Inputs	self, db_path
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Outputs	unspecified return value
Called by	-
Calls	threading.local, threading.RLock
Route	Memory/app/storage/sqlite_store.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as threading.RLock, threading.local.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

initialize

Function `initialize` executes domain-specific processing. Coordinates downstream calls such as conn.close, log.exception, self._configure_connection. Inputs: self. Outputs: bool. Called by: no internal callers detected. Calls: conn.close, log.exception, self._configure_connection.

Inputs	self
Outputs	bool
Called by	-
Calls	self._connect, self._configure_connection, self._init_db, conn.close, log.exception
Route	Memory/app/storage/sqlite_store.py::initialize
Tags	Utilities
Logic	Coordinates downstream calls such as conn.close, log.exception, self._configure_connection.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

is_available

Function `is\_available` executes domain-specific processing. Performs `is available` behavior for this module. Inputs: self. Outputs: bool. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self
Outputs	bool
Called by	-

Calls	-
Route	Memory/app/storage/sqlite_store.py::is_available
Tags	Utilities
Logic	Performs `is available` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`close_thread_connection`

Function `close\_thread\_connection` loads required project data. loads and retrieves required data Inputs: self. Outputs: no explicit return value. Called by: no internal callers detected. Calls: conn.close, getattr.

Inputs	self
Outputs	None
Called by	-
Calls	getattr, conn.close
Route	Memory/app/storage/sqlite_store.py::close_thread_connection
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`_connect`

Function `\_connect` executes domain-specific processing. Coordinates downstream calls such as sqlite3.connect. Inputs: self. Outputs: sqlite3.Connection. Called by: no internal callers detected. Calls: sqlite3.connect.

Inputs	self
Outputs	sqlite3.Connection
Called by	-
Calls	sqlite3.connect
Route	Memory/app/storage/sqlite_store.py::connect
Tags	Utilities
Logic	Coordinates downstream calls such as sqlite3.connect.
In Workflow	No

Workflow Reason	No entry-point-reachable call path detected for this function.
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_configure_connection

Function `\_configure\_connection` executes domain-specific processing. Coordinates downstream calls such as conn.commit, conn.execute. Inputs: self, conn. Outputs: no explicit return value. Called by: no internal callers detected. Calls: conn.commit, conn.execute.

Inputs	self, conn
Outputs	None
Called by	-
Calls	conn.execute, conn.commit
Route	Memory/app/storage/sqlite_store.py::configure_connection
Tags	Utilities
Logic	Coordinates downstream calls such as conn.commit, conn.execute.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_get_connection

Function `\_get\_connection` loads required project data. Retrieves module data required by downstream functions. Inputs: self. Outputs: sqlite3.Connection. Called by: no internal callers detected. Calls: RuntimeError, getattr, self._configure_connection.

Inputs	self
Outputs	sqlite3.Connection
Called by	-
Calls	self.initialize, RuntimeError, getattr, self._connect, self._configure_connection
Route	Memory/app/storage/sqlite_store.py::get_connection
Tags	Utilities
Logic	Retrieves module data required by downstream functions.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_execute_write

Function `\_execute\_write` persists outputs and artifacts. persists results to storage Inputs: self, query, params, commit. Outputs: sqlite3.Cursor. Called by: no internal callers detected. Calls: conn.commit, conn.execute, self._get_connection.

Inputs	self, query, params, commit
Outputs	sqlite3.Cursor
Called by	-
Calls	self._get_connection, conn.execute, conn.commit, str.lower, time.sleep
Route	Memory/app/storage/sqlite_store.py::execute_write
Tags	Utilities
Logic	persists results to storage
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_init_db

Function `\_init\_db` executes domain-specific processing. Coordinates downstream calls such as conn.commit, conn.execute, self._ensure_column. Inputs: self, conn. Outputs: no explicit return value. Called by: no internal callers detected. Calls: conn.commit, conn.execute, self._ensure_column.

Inputs	self, conn
Outputs	None
Called by	-
Calls	conn.execute, conn.commit, self._ensure_column
Route	Memory/app/storage/sqlite_store.py::init_db
Tags	Utilities
Logic	Coordinates downstream calls such as conn.commit, conn.execute, self._ensure_column.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_ensure_column

Function `\_ensure\_column` executes domain-specific processing. Coordinates downstream calls such as `conn.commit`, `conn.execute`, `conn.execute.fetchall`. Inputs: `self`, `conn`, `table`, `column`, `definition`. Outputs: no explicit return value. Called by: no internal callers detected. Calls: `conn.commit`, `conn.execute`, `conn.execute.fetchall`.

Inputs	<code>self</code> , <code>conn</code> , <code>table</code> , <code>column</code> , <code>definition</code>
Outputs	None
Called by	-
Calls	<code>conn.execute.fetchall</code> , <code>conn.execute</code> , <code>conn.commit</code>
Route	Memory/app/storage/sqlite_store.py::ensure_column
Tags	Utilities
Logic	Coordinates downstream calls such as <code>conn.commit</code> , <code>conn.execute</code> , <code>conn.execute.fetchall</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

add_memory

Function `add\_memory` executes domain-specific processing. Coordinates downstream calls such as `getattr`, `json.dumps`, `self._execute_write`. Inputs: `self`, `mem`. Outputs: `int`. Called by: no internal callers detected. Calls: `getattr`, `json.dumps`, `self._execute_write`.

Inputs	<code>self</code> , <code>mem</code>
Outputs	<code>int</code>
Called by	-
Calls	<code>self._execute_write</code> , <code>json.dumps</code> , <code>getattr</code>
Route	Memory/app/storage/sqlite_store.py::add_memory
Tags	Utilities
Logic	Coordinates downstream calls such as <code>getattr</code> , <code>json.dumps</code> , <code>self._execute_write</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

delete_memory

Function `delete\_memory` executes domain-specific processing. Coordinates downstream calls such as `self._execute_write`. Inputs: `self`, `memory_id`. Outputs: no explicit return value. Called by: no internal callers detected. Calls: `self._execute_write`.

Inputs	self, memory_id
Outputs	None
Called by	-
Calls	self._execute_write
Route	Memory/app/storage/sqlite_store.py::delete_memory
Tags	Utilities
Logic	Coordinates downstream calls such as self._execute_write.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

add_interaction

Function `add\_interaction` executes domain-specific processing. Coordinates downstream calls such as json.dumps, self._execute_write. Inputs: self, interaction. Outputs: int. Called by: no internal callers detected. Calls: json.dumps, self._execute_write.

Inputs	self, interaction
Outputs	int
Called by	-
Calls	self._execute_write, json.dumps
Route	Memory/app/storage/sqlite_store.py::add_interaction
Tags	Utilities
Logic	Coordinates downstream calls such as json.dumps, self._execute_write.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

update_status

Function `update\_status` executes domain-specific processing. Coordinates downstream calls such as self._execute_write. Inputs: self, memory_id, status. Outputs: no explicit return value. Called by: no internal callers detected. Calls: self._execute_write.

Inputs	self, memory_id, status
Outputs	None
Called by	-

Calls	self._execute_write
Route	Memory/app/storage/sqlite_store.py::update_status
Tags	Utilities
Logic	Coordinates downstream calls such as self._execute_write.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

get_memory

Function `get\_memory` loads required project data. Retrieves module data required by downstream functions. Inputs: self, memory_id. Outputs: Optional. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchone, self._get_connection.

Inputs	self, memory_id
Outputs	Optional
Called by	-
Calls	self._get_connection, conn.execute.fetchone, conn.execute, self._row_to_memory
Route	Memory/app/storage/sqlite_store.py::get_memory
Tags	Utilities
Logic	Retrieves module data required by downstream functions.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

search

Function `search` serves application-facing request handling. Processes `search` request flow and returns structured response data. Inputs: self, text, limit. Outputs: List. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchall, self._get_connection.

Inputs	self, text, limit
Outputs	List
Called by	-
Calls	self._get_connection, conn.execute.fetchall, conn.execute, self._row_to_memory

Route	Memory/app/storage/sqlite_store.py::search
Tags	Retrieval, Workflow
Logic	Processes `search` request flow and returns structured response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

list_recent

Function `list\_recent` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: self, limit. Outputs: List. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchall, self._get_connection.

Inputs	self, limit
Outputs	List
Called by	-
Calls	self._get_connection, conn.execute.fetchall, conn.execute, self._row_to_memory
Route	Memory/app/storage/sqlite_store.py::list_recent
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

list_recent_interactions

Function `list\_recent\_interactions` serves application-facing request handling. Retrieve recent interactions for deduplication checking. Inputs: self, limit, minutes. Outputs: List. Called by: no internal callers detected. Calls: InteractionMetadata, conn.execute, conn.execute.fetchall.

Inputs	self, limit, minutes
Outputs	List
Called by	-
Calls	self._get_connection, conn.execute.fetchall, conn.execute, interactions.append, InteractionMetadata, json.loads
Route	Memory/app/storage/sqlite_store.py::list_recent_interactions
Tags	Workflow

Logic	Retrieve recent interactions for deduplication checking.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

add_edge

Function `add\_edge` executes domain-specific processing. Coordinates downstream calls such as `self._execute_write`. Inputs: `self`, `from_id`, `to_id`, `relation`, `weight`. Outputs: `bool`. Called by: no internal callers detected. Calls: `self._execute_write`.

Inputs	<code>self</code> , <code>from_id</code> , <code>to_id</code> , <code>relation</code> , <code>weight</code>
Outputs	<code>bool</code>
Called by	-
Calls	<code>self._execute_write</code>
Route	<code>Memory/app/storage/sqlite_store.py::add_edge</code>
Tags	Tagging
Logic	Coordinates downstream calls such as <code>self._execute_write</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

get_edges

Function `get\_edges` loads required project data. Retrieves module data required by downstream functions. Inputs: `self`, `memory_id`. Outputs: `List`. Called by: no internal callers detected. Calls: `conn.execute`, `conn.execute.fetchall`, `self._get_connection`.

Inputs	<code>self</code> , <code>memory_id</code>
Outputs	<code>List</code>
Called by	-
Calls	<code>self._get_connection</code> , <code>conn.execute.fetchall</code> , <code>conn.execute</code>
Route	<code>Memory/app/storage/sqlite_store.py::get_edges</code>
Tags	Tagging
Logic	Retrieves module data required by downstream functions.
In Workflow	No

Workflow Reason	No entry-point-reachable call path detected for this function.
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list_interactions

Function `list\_interactions` serves application-facing request handling. Processes `list interactions` request flow and returns structured response data. Inputs: self, limit. Outputs: List. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchall, self._get_connection.

Inputs	self, limit
Outputs	List
Called by	-
Calls	self._get_connection, conn.execute.fetchall, conn.execute, self._row_to_interaction
Route	Memory/app/storage/sqlite_store.py::list_interactions
Tags	Utilities
Logic	Processes `list interactions` request flow and returns structured response data.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

interaction_stats

Function `interaction\_stats` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: self, limit. Outputs: Dict. Called by: no internal callers detected. Calls: self.list_interactions.

Inputs	self, limit
Outputs	Dict
Called by	-
Calls	self.list_interactions
Route	Memory/app/storage/sqlite_store.py::interaction_stats
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

prune_interactions

Function `prune\_interactions` orchestrates execution flow. orchestrates workflow execution steps
 Inputs: self, retention_days, max_rows, batch_size. Outputs: int. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchone, cutoff.strftime.

Inputs	self, retention_days, max_rows, batch_size
Outputs	int
Called by	-
Calls	datetime.now, timedelta, self._execute_write, cutoff.strftime, self._get_connection, conn.execute.fetchone, conn.execute
Route	Memory/app/storage/sqlite_store.py::prune_interactions
Tags	Workflow
Logic	orchestrates workflow execution steps
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

prune_edges

Function `prune\_edges` orchestrates execution flow. orchestrates workflow execution steps
 Inputs: self, retention_days, max_rows, batch_size. Outputs: int. Called by: no internal callers detected. Calls: conn.execute, conn.execute.fetchone, cutoff.strftime.

Inputs	self, retention_days, max_rows, batch_size
Outputs	int
Called by	-
Calls	datetime.now, timedelta, self._execute_write, cutoff.strftime, self._get_connection, conn.execute.fetchone, conn.execute
Route	Memory/app/storage/sqlite_store.py::prune_edges
Tags	Workflow
Logic	orchestrates workflow execution steps
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

_row_to_memory

Function `\_row\_to\_memory` executes domain-specific processing. Coordinates downstream calls such as MemoryNode, MemoryStatus, MemoryType. Inputs: self, row. Outputs: MemoryNode. Called by: no internal callers detected. Calls: MemoryNode, MemoryStatus, MemoryType.

Inputs	self, row
Outputs	MemoryNode
Called by	-
Calls	MemoryNode, MemoryType, json.loads, row.keys, MemoryStatus
Route	Memory/app/storage/sqlite_store.py::row_to_memory
Tags	Utilities
Logic	Coordinates downstream calls such as MemoryNode, MemoryStatus, MemoryType.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_row_to_interaction

Function `\_row\_to\_interaction` executes domain-specific processing. Coordinates downstream calls such as json.loads. Inputs: self, row. Outputs: Dict. Called by: no internal callers detected. Calls: json.loads.

Inputs	self, row
Outputs	Dict
Called by	-
Calls	json.loads
Route	Memory/app/storage/sqlite_store.py::row_to_interaction
Tags	Utilities
Logic	Coordinates downstream calls such as json.loads.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as warnings.warn. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: warnings.warn.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	warnings.warn
Route	Memory/app/storage/vector_store.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as warnings.warn.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

upsert

Function `upsert` executes domain-specific processing. Coordinates downstream calls such as embedder.embed. Inputs: self, item_id, text. Outputs: no explicit return value. Called by: no internal callers detected. Calls: embedder.embed.

Inputs	self, item_id, text
Outputs	None
Called by	-
Calls	self._items.append, embedder.embed
Route	Memory/app/storage/vector_store.py::upsert
Tags	Utilities
Logic	Coordinates downstream calls such as embedder.embed.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

search

Function `search` serves application-facing request handling. Processes `search` request flow and returns structured response data. Inputs: self, query, top_k. Outputs: List. Called by: no internal callers detected. Calls: cosine, embedder.embed, scored.sort.

Inputs	self, query, top_k
Outputs	List
Called by	-

Calls	embedder.embed, cosine, scored.sort
Route	Memory/app/storage/vector_store.py::search
Tags	Retrieval, Workflow
Logic	Processes `search` request flow and returns structured response data.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

cosine

Function `cosine` executes domain-specific processing. Coordinates downstream calls such as `math.sqrt`, `zip`. Inputs: `a`, `b`. Outputs: `float`. Called by: `search`. Calls: `math.sqrt`, `zip`.

Inputs	<code>a</code> , <code>b</code>
Outputs	<code>float</code>
Called by	<code>search</code>
Calls	<code>zip</code> , <code>math.sqrt</code>
Route	Memory/app/storage/vector_store.py::cosine
Tags	Utilities
Logic	Coordinates downstream calls such as <code>math.sqrt</code> , <code>zip</code> .
In Workflow	Yes
Workflow Reason	Reachable from entry points and actively participates in call flow.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as `AttentionEngine`, `ContextManager`, `FallbackManager`. Inputs: `self`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `AttentionEngine`, `ContextManager`, `FallbackManager`.

Inputs	<code>self</code>
Outputs	unspecified return value
Called by	-

Calls	AttentionEngine, QueryRouter, FallbackManager, HybridRetriever, ContextManager, MemoryExtractor, MemoryValidator, MemoryLifecycle, TreeIndexer, InteractionFilter
Route	Memory/app/workflow/orchestrator.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as AttentionEngine, ContextManager, FallbackManager.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

run

Function `run` orchestrates execution flow. orchestrates workflow execution steps Inputs: self, user_id, session_id, query, debug. Outputs: QueryResult. Called by: no internal callers detected. Calls: IntentType, QueryResult, llm.chat.

Inputs	self, user_id, session_id, query, debug
Outputs	QueryResult
Called by	-
Calls	self.attention.analyze, self.router.route, self._log_interaction, IntentType, self.router.policies.get_policy, merged_domains.update, self.retriever.retrieve, self.context.build_context, self.fallback.apply_fallback, llm.chat, self.extractor.extract, self.validator.validate, self.lifecycle.reinforce, memory_service.store, self._build_graph_edges, sqlite_store.update_status, sqlite_store.list_recent, self.tree_indexer.build_from_memories, persistent_tree_store.save, QueryResult
Route	Memory/app/workflow/orchestrator.py::run
Tags	Workflow
Logic	orchestrates workflow execution steps
In Workflow	Yes

Workflow Reason	Reachable from entry points and dispatches work to downstream functions.
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_log_interaction

Function `\_log\_interaction` executes domain-specific processing. Coordinates downstream calls such as InteractionMetadata, self.interaction_filter.should_store, sqlite_store.add_interaction. Inputs: self, user_id, session_id, query, route, state. Outputs: computed value. Called by: no internal callers detected. Calls: InteractionMetadata, self.interaction_filter.should_store, sqlite_store.add_interaction.

Inputs	self, user_id, session_id, query, route, state
Outputs	computed value
Called by	-
Calls	InteractionMetadata, sqlite_store.list_recent_interactions, self.interaction_filter.should_store, sqlite_store.add_interaction
Route	Memory/app/workflow/orchestrator.py::log_interaction
Tags	Utilities
Logic	Coordinates downstream calls such as InteractionMetadata, self.interaction_filter.should_store, sqlite_store.add_interaction.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

_build_graph_edges

Function `\_build\_graph\_edges` builds and composes output artifacts. Create real graph edges: Inputs: self, stored_ids, memories. Outputs: no explicit return value. Called by: no internal callers detected. Calls: graph_store.add_relation, graph_store.connect_related_memories, zip.

Inputs	self, stored_ids, memories
Outputs	None
Called by	-
Calls	zip, graph_ids.append, other_ids.append, graph_store.add_relation, graph_store.connect_related_memories
Route	Memory/app/workflow/orchestrator.py::build_graph_edges
Tags	Tagging

Logic	Create real graph edges:
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

analyze_emotional_signals

Function `analyze\_emotional\_signals` executes domain-specific processing. Run all signal layers and produce an EmotionalSignals object. Inputs: text. Outputs: EmotionalSignals. Called by: no internal callers detected. Calls: EmotionalSignals, _AGGRESSION_PHRASES.search, _ALL_CAPS_WORD.findall.

Inputs	text
Outputs	EmotionalSignals
Called by	-
Calls	text.lower.strip, text.lower, re.findall, _MULTI_EXCLAIM.search, punct_reasons.append, _MULTI_QUESTION.search, _INTERROBANG.search, _ALL_CAPS_WORD.findall, _AGGRESSION_PHRASES.search, _GRATITUDE_PHRASES.search, frust_reasons.append, _EXPLICIT_CONFUSION.search, conf_reasons.append, urg_reasons.append, agg_reasons.append, pos_reasons.append, round, EmotionalSignals
Route	Memory/app/memory/behavior/emotional_analyzer.py::analyze_emotional_signals
Tags	Utilities
Logic	Run all signal layers and produce an EmotionalSignals object.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_similarity

Function `\_similarity` executes domain-specific processing. Token-level Jaccard similarity. Inputs: self, a, b. Outputs: float. Called by: no internal callers detected. Calls: a.lower.split, b.lower.split.

Inputs	self, a, b
Outputs	float
Called by	-
Calls	a.lower.split, a.lower, b.lower.split, b.lower
Route	Memory/app/memory/consolidation/compaction_engine.py::similarity
Tags	Utilities

Logic	Token-level Jaccard similarity.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`_group_by_topic`

Function ``_group_by_topic`` executes domain-specific processing. Group memories by `memory_type`, then cluster similar ones. Inputs: `self`, `memories`. Outputs: `dict`. Called by: no internal callers detected. Calls: `defaultdict`.

Inputs	<code>self</code> , <code>memories</code>
Outputs	<code>dict</code>
Called by	-
Calls	<code>defaultdict</code> , <code>groups.append</code>
Route	<code>Memory/app/memory/consolidation/compaction_engine.py::group_by_topic</code>
Tags	Utilities
Logic	Group memories by <code>memory_type</code> , then cluster similar ones.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`compact`

Function ``compact`` executes domain-specific processing. Coordinates downstream calls such as `MemoryNode`, `assigned.add`, `dict.fromkeys`. Inputs: `self`, `memories`. Outputs: `List`. Called by: no internal callers detected. Calls: `MemoryNode`, `assigned.add`, `dict.fromkeys`.

Inputs	<code>self</code> , <code>memories</code>
Outputs	<code>List</code>
Called by	-
Calls	<code>self._group_by_topic</code> , <code>groups.items</code> , <code>assigned.add</code> , <code>self._similarity</code> , <code>cluster.append</code> , <code>clusters.append</code> , <code>results.append</code> , <code>m.content.strip.lower</code> , <code>m.content.strip</code> , <code>seen_snippets.add</code> , <code>unique_content.append</code> , <code>MemoryNode</code> , <code>dict.fromkeys</code> , <code>getattr</code>
Route	<code>Memory/app/memory/consolidation/compaction_engine.py::compact</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>MemoryNode</code> , <code>assigned.add</code> , <code>dict.fromkeys</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

decay

Function `decay` executes domain-specific processing. Performs `decay` behavior for this module. Inputs: self, memories. Outputs: List. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, memories
Outputs	List
Called by	-
Calls	-
Route	Memory/app/memory/consolidation/memory_lifecycle.py::decay
Tags	Utilities
Logic	Performs `decay` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

reinforce

Function `reinforce` executes domain-specific processing. Performs `reinforce` behavior for this module. Inputs: self, memories. Outputs: List. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, memories
Outputs	List
Called by	-
Calls	-
Route	Memory/app/memory/consolidation/memory_lifecycle.py::reinforce
Tags	Utilities
Logic	Performs `reinforce` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

reflect

Function `reflect` executes domain-specific processing. Coordinates downstream calls such as Counter, MemoryNode, entity_counter.most_common. Inputs: self, memories. Outputs: List. Called by: no internal callers detected. Calls: Counter, MemoryNode, entity_counter.most_common.

Inputs	self, memories
Outputs	List

Called by	-
Calls	Counter, m.content.lower.split, m.content.lower, word_counter.update, word_counter.most_common, reflections.append, MemoryNode, entity_counter.update, entity_counter.most_common, m.content.split
Route	Memory/app/memory/consolidation/reflect_engine.py::reflect
Tags	Utilities
Logic	Coordinates downstream calls such as Counter, MemoryNode, entity_counter.most_common.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

consolidate

Function `consolidate` executes domain-specific processing. Performs `consolidate` behavior for this module. Inputs: self, memories. Outputs: List. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, memories
Outputs	List
Called by	-
Calls	-
Route	Memory/app/memory/consolidation/sleep_cycle.py::consolidate
Tags	Utilities
Logic	Performs `consolidate` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_tokenize

Function `\_tokenize` executes domain-specific processing. parses and extracts structured information Inputs: self, text. Outputs: set. Called by: no internal callers detected. Calls: text.lower.split.

Inputs	self, text
Outputs	set
Called by	-
Calls	text.lower.split, text.lower
Route	Memory/app/memory/extraction/contradiction_detector.py::tokenize

Tags	Utilities
Logic	parses and extracts structured information
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_jaccard

Function `\_jaccard` executes domain-specific processing. Coordinates downstream calls such as `self._tokenize`. Inputs: `self`, `a`, `b`. Outputs: `float`. Called by: no internal callers detected. Calls: `self._tokenize`.

Inputs	<code>self</code> , <code>a</code> , <code>b</code>
Outputs	<code>float</code>
Called by	-
Calls	<code>self._tokenize</code>
Route	<code>Memory/app/memory/extraction/contradiction_detector.py::jaccard</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>self._tokenize</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_has_negation_flip

Function `\_has\_negation\_flip` executes domain-specific processing. Check if one text negates the other. Inputs: `self`, `text_a`, `text_b`. Outputs: `bool`. Called by: no internal callers detected. Calls: `self._tokenize`.

Inputs	<code>self</code> , <code>text_a</code> , <code>text_b</code>
Outputs	<code>bool</code>
Called by	-
Calls	<code>self._tokenize</code>
Route	<code>Memory/app/memory/extraction/contradiction_detector.py::has_negation_flip</code>
Tags	Utilities
Logic	Check if one text negates the other.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

find_contradictions

Function ``find_contradictions`` executes domain-specific processing. Returns list of (existing_id, new_id) pairs where existing Inputs: self, new_memories, existing. Outputs: List. Called by: no internal callers detected. Calls: self._has_negation_flip, self._jaccard.

Inputs	self, new_memories, existing
Outputs	List
Called by	-
Calls	self._jaccard, self._has_negation_flip, contradictions.append
Route	Memory/app/memory/extraction/contradiction_detector.py::find_contradictions
Tags	Utilities
Logic	Returns list of (existing_id, new_id) pairs where existing
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

``_split_atomic_facts``

Function ``_split_atomic_facts`` executes domain-specific processing. Break a user utterance into individual factual claims. Inputs: text. Outputs: List. Called by: no internal callers detected. Calls: re.split, seen.add.

Inputs	text
Outputs	List
Called by	-
Calls	re.split, p.strip, f.lower, seen.add, unique.append, text.strip
Route	Memory/app/memory/extraction/memory_extractor.py::split_atomic_facts
Tags	Utilities
Logic	Break a user utterance into individual factual claims.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

``_compress_episode``

Function ``_compress_episode`` executes domain-specific processing. Create a ≤ 2 sentence event summary instead of storing raw output. Inputs: user_text, assistant_text. Outputs: str. Called by: no internal callers detected. Calls: assistant_text.split.

Inputs	user_text, assistant_text
Outputs	str
Called by	-

Calls	user_text.strip, assistant_text.split.strip, assistant_text.split
Route	Memory/app/memory/extraction/memory_extractor.py::compress_episode
Tags	Utilities
Logic	Create a ≤ 2 sentence event summary instead of storing raw output.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_extract_entities

Function `\_extract\_entities` extracts and structures data. Pull capitalised words / tech terms as entity candidates. Inputs: text. Outputs: List. Called by: no internal callers detected. Calls: dict.fromkeys, re.findall, t.title.

Inputs	text
Outputs	List
Called by	-
Calls	re.findall, dict.fromkeys, t.title
Route	Memory/app/memory/extraction/memory_extractor.py::extract_entities
Tags	Parsing
Logic	Pull capitalised words / tech terms as entity candidates.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_compute_quality

Function `\_compute\_quality` executes domain-specific processing. Coordinates downstream calls such as c.isupper, round. Inputs: text, importance. Outputs: float. Called by: no internal callers detected. Calls: c.isupper, round.

Inputs	text, importance
Outputs	float
Called by	-
Calls	c.isupper, round
Route	Memory/app/memory/extraction/memory_extractor.py::compute_quality
Tags	Utilities
Logic	Coordinates downstream calls such as c.isupper, round.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_extract_relations

Function `\_extract\_relations` extracts and structures data. Extract (subject, predicate, object) triples from text. Inputs: text. Outputs: List. Called by: no internal callers detected. Calls: builder, m.group, re.finditer.

Inputs	text
Outputs	List
Called by	-
Calls	m.group, m.group.strip, re.finditer, builder, s.strip, triples.append
Route	Memory/app/memory/extraction/memory_extractor.py::extract_relations
Tags	Parsing
Logic	Extract (subject, predicate, object) triples from text.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

extract

Function `extract` extracts and structures data. parses and extracts structured information Inputs: self, user_text, assistant_text, state. Outputs: ExtractionResult. Called by: no internal callers detected. Calls: ExtractionResult, MemoryNode, self._compress_episode.

Inputs	self, user_text, assistant_text, state
Outputs	ExtractionResult
Called by	-
Calls	self._extract_entities, self._split_atomic_facts, fact.lower, semantic.append, MemoryNode, self._compute_quality, self._compress_episode, self._extract_relations, graph.append, signals.append, reflective.append, ExtractionResult
Route	Memory/app/memory/extraction/memory_extractor.py::extract
Tags	Parsing
Logic	parses and extracts structured information
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as ContradictionDetector. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: ContradictionDetector.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	ContradictionDetector
Route	Memory/app/memory/extraction/memory_validator.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as ContradictionDetector.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_tokenize

Function `\_tokenize` executes domain-specific processing. parses and extracts structured information Inputs: self, text. Outputs: set. Called by: no internal callers detected. Calls: text.strip.lower.split.

Inputs	self, text
Outputs	set
Called by	-
Calls	text.strip.lower.split, text.strip.lower, text.strip
Route	Memory/app/memory/extraction/memory_validator.py::tokenize
Tags	Utilities
Logic	parses and extracts structured information
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_similarity

Function `\_similarity` executes domain-specific processing. Coordinates downstream calls such as self._tokenize. Inputs: self, a, b. Outputs: float. Called by: no internal callers detected. Calls: self._tokenize.

Inputs	self, a, b
Outputs	float
Called by	-
Calls	self._tokenize
Route	Memory/app/memory/extraction/memory_validator.py::similarity

Tags	Utilities
Logic	Coordinates downstream calls such as self._tokenize.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

validate

Function `validate` validates constraints and expected state. validates inputs and enforces constraints Inputs: self, new_memories, existing_memories. Outputs: Tuple. Called by: no internal callers detected. Calls: self._similarity, self.detector.find_contradictions.

Inputs	self, new_memories, existing_memories
Outputs	Tuple
Called by	-
Calls	self.detector.find_contradictions, invalidations.append, mem.content.strip, quality_filtered.append, self._similarity, deduped.append
Route	Memory/app/memory/extraction/memory_validator.py::validate
Tags	Workflow
Logic	validates inputs and enforces constraints
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

_phrase_score

Function `\_phrase\_score` executes domain-specific processing. Score based on phrase matching with logarithmic stacking. Inputs: text, strong, moderate, strong_weight, moderate_weight. Outputs: float. Called by: analyze. Calls: no internal downstream calls.

Inputs	text, strong, moderate, strong_weight, moderate_weight
Outputs	float
Called by	analyze
Calls	-
Route	Memory/app/memory/orchestration/attention_engine.py::phrase_score
Tags	Utilities
Logic	Score based on phrase matching with logarithmic stacking.
In Workflow	Yes
Workflow Reason	Reachable from entry points as part of execution path.

analyze

Function `analyze` executes domain-specific processing. Coordinates downstream calls such as BehaviorState, _phrase_score, analyze_emotional_signals. Inputs: self, query. Outputs: BehaviorState. Called by: no internal callers detected. Calls: BehaviorState, _phrase_score, analyze_emotional_signals.

Inputs	self, query
Outputs	BehaviorState
Called by	-
Calls	query.lower.strip, query.lower, analyze_emotional_signals, _phrase_score, q.count, INTENT_KEYWORDS.items, BehaviorState, round
Route	Memory/app/memory/orchestration/attention_engine.py::analyze
Tags	Utilities
Logic	Coordinates downstream calls such as BehaviorState, _phrase_score, analyze_emotional_signals.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

decide

Function `decide` executes domain-specific processing. Performs `decide` behavior for this module. Inputs: self, route_flags. Outputs: str. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, route_flags
Outputs	str
Called by	-
Calls	route_flags.items
Route	Memory/app/memory/orchestration/decision_engine.py::decide
Tags	Utilities
Logic	Performs `decide` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

segment

Function `segment` executes domain-specific processing. Performs `segment` behavior for this module. Inputs: self, memories. Outputs: List. Called by: no internal callers detected. Calls: no

internal downstream calls.

Inputs	self, memories
Outputs	List
Called by	-
Calls	current.append, episodes.append
Route	Memory/app/memory/orchestration/episode_engine.py::segment
Tags	Utilities
Logic	Performs `segment` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Args: Inputs: self, lookback_minutes. Outputs: unspecified return value. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, lookback_minutes
Outputs	unspecified return value
Called by	-
Calls	-
Route	Memory/app/memory/orchestration/interaction_filter.py::init
Tags	Utilities
Logic	Args:
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_normalize_query

Function `\_normalize\_query` filters and normalizes intermediate data. Normalize query for comparison. Inputs: self, query. Outputs: str. Called by: no internal callers detected. Calls: re.sub.

Inputs	self, query
Outputs	str
Called by	-
Calls	query.lower.strip, query.lower, re.sub
Route	Memory/app/memory/orchestration/interaction_filter.py::normalize_query

Tags	Utilities
Logic	Normalize query for comparison.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_compute_query_hash

Function `\_compute\_query\_hash` executes domain-specific processing. Compute hash of normalized query for fast comparison. Inputs: self, query. Outputs: str. Called by: no internal callers detected. Calls: hashlib.md5, hashlib.md5.hexdigest, normalized.encode.

Inputs	self, query
Outputs	str
Called by	-
Calls	self._normalize_query, hashlib.md5.hexdigest, hashlib.md5, normalized.encode
Route	Memory/app/memory/orchestration/interaction_filter.py::compute_query_hash
Tags	Utilities
Logic	Compute hash of normalized query for fast comparison.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_compute_token_overlap

Function `\_compute\_token\_overlap` executes domain-specific processing. Compute Jaccard similarity between token sets. Inputs: self, query1, query2. Outputs: float. Called by: no internal callers detected. Calls: self._normalize_query.split, self._normalize_query.split.

Inputs	self, query1, query2
Outputs	float
Called by	-
Calls	self._normalize_query.split, self._normalize_query
Route	Memory/app/memory/orchestration/interaction_filter.py::compute_token_overlap
Tags	Utilities
Logic	Compute Jaccard similarity between token sets.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`_score_importance`

Function `_score_importance` executes domain-specific processing. Score interaction importance (0.0 to 1.0). Inputs: self, interaction. Outputs: float. Called by: no internal callers detected. Calls: re.search.

Inputs	self, interaction
Outputs	float
Called by	-
Calls	interaction.query_fragment.lower, re.search
Route	Memory/app/memory/orchestration/interaction_filter.py::score_importance
Tags	Utilities
Logic	Score interaction importance (0.0 to 1.0).
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`_check_duplicate`

Function `_check_duplicate` validates constraints and expected state. Check if interaction is a duplicate of recent interactions. Inputs: self, interaction, recent_interactions. Outputs: Tuple. Called by: no internal callers detected. Calls: self._compute_query_hash, self._compute_token_overlap.

Inputs	self, interaction, recent_interactions
Outputs	Tuple
Called by	-
Calls	self._compute_query_hash, self._compute_token_overlap
Route	Memory/app/memory/orchestration/interaction_filter.py::check_duplicate
Tags	Utilities
Logic	Check if interaction is a duplicate of recent interactions.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

`should_store`

Function `should_store` executes domain-specific processing. Determine if interaction should be stored. Inputs: self, interaction, recent_interactions. Outputs: Tuple. Called by: no internal callers detected. Calls: self._check_duplicate, self._score_importance.

Inputs	self, interaction, recent_interactions
Outputs	Tuple

Called by	-
Calls	self._score_importance, self._check_duplicate
Route	Memory/app/memory/orchestration/interaction_filter.py::should_store
Tags	Utilities
Logic	Determine if interaction should be stored.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

build_context

Function `build\_context` builds and composes output artifacts. Builds structured artifacts consumed by downstream stages. Inputs: self, memories, max_chars. Outputs: str. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, memories, max_chars
Outputs	str
Called by	-
Calls	m.memory_type.value.upper, parts.append
Route	Memory/app/memory/retrieval/context_manager.py::build_context
Tags	Retrieval
Logic	Builds structured artifacts consumed by downstream stages.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as TreeNode. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: TreeNode.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	TreeNode
Route	Memory/app/memory/retrieval/context_tree.py::init
Tags	Retrieval
Logic	Coordinates downstream calls such as TreeNode.
In Workflow	No

Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.
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add_path

Function `add\_path` executes domain-specific processing. Coordinates downstream calls such as `TreeNode`, `next`. Inputs: `self`, `titles`, `summary`, `raw_text`. Outputs: no explicit return value. Called by: no internal callers detected. Calls: `TreeNode`, `next`.

Inputs	<code>self</code> , <code>titles</code> , <code>summary</code> , <code>raw_text</code>
Outputs	None
Called by	-
Calls	<code>next</code> , <code>TreeNode</code> , <code>node.children.append</code>
Route	<code>Memory/app/memory/retrieval/context_tree.py::add_path</code>
Tags	Retrieval
Logic	Coordinates downstream calls such as <code>TreeNode</code> , <code>next</code> .
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

to_dict

Function `to\_dict` executes domain-specific processing. Coordinates downstream calls such as `walk`. Inputs: `self`. Outputs: `dict`. Called by: no internal callers detected. Calls: `walk`.

Inputs	<code>self</code>
Outputs	<code>dict</code>
Called by	-
Calls	<code>walk</code>
Route	<code>Memory/app/memory/retrieval/context_tree.py::to_dict</code>
Tags	Retrieval
Logic	Coordinates downstream calls such as <code>walk</code> .
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

walk

Function `walk` executes domain-specific processing. Coordinates downstream calls such as `walk`. Inputs: `n`. Outputs: `dict`. Called by: `to_dict`, `walk`. Calls: `walk`.

Inputs	n
Outputs	dict
Called by	to_dict, walk
Calls	walk
Route	Memory/app/memory/retrieval/context_tree.py::walk
Tags	Retrieval
Logic	Coordinates downstream calls such as walk.
In Workflow	Yes
Workflow Reason	Reachable from entry points and actively participates in call flow.

summarize

Function `summarize` executes domain-specific processing. Performs `summarize` behavior for this module. Inputs: self, memories. Outputs: str. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, memories
Outputs	str
Called by	-
Calls	lines.append
Route	Memory/app/memory/retrieval/hierarchical_summarizer.py::summarize
Tags	Retrieval
Logic	Performs `summarize` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as RetrievalRanker. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: RetrievalRanker.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	RetrievalRanker
Route	Memory/app/memory/retrieval/hybrid_retriever.py::init

Tags	Retrieval
Logic	Coordinates downstream calls such as RetrievalRanker.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

retrieve

Function `retrieve` executes domain-specific processing. loads and retrieves required data

Inputs: self, query, route_decision, behavior, with_audit. Outputs: computed value. Called by: no internal callers detected. Calls: TreeRetriever, faiss_store.search, graph_store.neighborhood.

Inputs	self, query, route_decision, behavior, with_audit
Outputs	computed value
Called by	-
Calls	sqlite_store.search, faiss_store.search, sqlite_store.get_memory, merged.values, graph_store.neighborhood, persistent_tree_store.load, TreeRetriever, tree_retriever.retrieve, sqlite_store.list_recent, self.ranker.rank
Route	Memory/app/memory/retrieval/hybrid_retriever.py::retrieve
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as Path. Inputs: self, path. Outputs: unspecified return value. Called by: no internal callers detected. Calls: Path.

Inputs	self, path
Outputs	unspecified return value
Called by	-
Calls	Path
Route	Memory/app/memory/retrieval/persistent_tree_store.py::init
Tags	Retrieval

Logic	Coordinates downstream calls such as Path.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

save

Function `save` persists outputs and artifacts. persists results to storage
 Inputs: self, tree.
 Outputs: unspecified return value. Called by: no internal callers detected. Calls: json.dump, open, tree.to_dict.

Inputs	self, tree
Outputs	unspecified return value
Called by	-
Calls	open, json.dump, tree.to_dict
Route	Memory/app/memory/retrieval/persistent_tree_store.py::save
Tags	Retrieval
Logic	persists results to storage
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

load

Function `load` loads required project data. loads and retrieves required data
 Inputs: self.
 Outputs: ContextTree. Called by: no internal callers detected. Calls: ContextTree, json.load, open.

Inputs	self
Outputs	ContextTree
Called by	-
Calls	ContextTree, self.path.exists, open, json.load, rebuild
Route	Memory/app/memory/retrieval/persistent_tree_store.py::load
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

rebuild

Function `rebuild` builds and composes output artifacts. Builds structured artifacts consumed by downstream stages. Inputs: node_data. Outputs: unspecified return value. Called by: load, rebuild. Calls: TreeNode, rebuild.

Inputs	node_data
Outputs	unspecified return value
Called by	load, rebuild
Calls	TreeNode, node.children.append, rebuild
Route	Memory/app/memory/retrieval/persistent_tree_store.py::rebuild
Tags	Retrieval
Logic	Builds structured artifacts consumed by downstream stages.
In Workflow	Yes
Workflow Reason	Reachable from entry points and actively participates in call flow.

resolve

Function `resolve` executes domain-specific processing. Performs `resolve` behavior for this module. Inputs: self, query. Outputs: dict. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, query
Outputs	dict
Called by	-
Calls	query.lower
Route	Memory/app/memory/retrieval/query_resolver.py::resolve
Tags	Retrieval, Parsing
Logic	Performs `resolve` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

score

Function `score` executes domain-specific processing. Coordinates downstream calls such as getattr, query.lower.split, round. Inputs: self, query, memory, behavior. Outputs: float. Called by: no internal callers detected. Calls: getattr, query.lower.split, round.

Inputs	self, query, memory, behavior
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Outputs	float
Called by	-
Calls	query.lower.split, query.lower, getattr, round
Route	Memory/app/memory/retrieval/retrieval_ranker.py::score
Tags	Retrieval
Logic	Coordinates downstream calls such as getattr, query.lower.split, round.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

rank

Function `rank` executes domain-specific processing. Coordinates downstream calls such as scored.sort, seen.add, self.score. Inputs: self, query, memories, top_k, behavior. Outputs: List. Called by: no internal callers detected. Calls: scored.sort, seen.add, self.score.

Inputs	self, query, memories, top_k, behavior
Outputs	List
Called by	-
Calls	m.content.strip.lower, m.content.strip, seen.add, unique.append, self.score, scored.sort
Route	Memory/app/memory/retrieval/retrieval_ranker.py::rank
Tags	Retrieval
Logic	Coordinates downstream calls such as scored.sort, seen.add, self.score.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_classify

Function `\_classify` executes domain-specific processing. Classify into (topic, subtopic) by keyword matching. Inputs: self, text. Outputs: Tuple. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, text
Outputs	Tuple
Called by	-
Calls	text.lower, TOPIC_TAXONOMY.items, subtopics.items

Route	Memory/app/memory/retrieval/tree_indexer.py::classify
Tags	Retrieval
Logic	Classify into (topic, subtopic) by keyword matching.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_generate_branch_summary

Function `\_generate\_branch\_summary` builds and composes output artifacts. Create a compressed summary of memories in a branch. Inputs: self, mems. Outputs: str. Called by: no internal callers detected. Calls: Counter, entity_freq.most_common, entity_freq.update.

Inputs	self, mems
Outputs	str
Called by	-
Calls	Counter, entity_freq.update, entity_freq.most_common
Route	Memory/app/memory/retrieval/tree_indexer.py::generate_branch_summary
Tags	Retrieval
Logic	Create a compressed summary of memories in a branch.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

build_from_memories

Function `build\_from\_memories` builds and composes output artifacts. Builds structured artifacts consumed by downstream stages. Inputs: self, memories. Outputs: ContextTree. Called by: no internal callers detected. Calls: ContextTree, defaultdict, self._classify.

Inputs	self, memories
Outputs	ContextTree
Called by	-
Calls	ContextTree, defaultdict, self._classify, classified.append, classified.items, subtopics.values, self._generate_branch_summary, tree.add_path, subtopics.items, classified.values
Route	Memory/app/memory/retrieval/tree_indexer.py::build_from_memories
Tags	Retrieval
Logic	Builds structured artifacts consumed by downstream stages.
In Workflow	No

Workflow Reason	No entry-point-reachable call path detected for this function.
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__init__

Function `__init__` executes domain-specific processing. Coordinates downstream calls such as ContextTree. Inputs: self, tree. Outputs: unspecified return value. Called by: no internal callers detected. Calls: ContextTree.

Inputs	self, tree
Outputs	unspecified return value
Called by	-
Calls	ContextTree
Route	Memory/app/memory/retrieval/tree_retriever.py::init
Tags	Retrieval
Logic	Coordinates downstream calls such as ContextTree.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

_score_node

Function `_score_node` executes domain-specific processing. Coordinates downstream calls such as q.split. Inputs: self, query, node. Outputs: float. Called by: no internal callers detected. Calls: q.split.

Inputs	self, query, node
Outputs	float
Called by	-
Calls	query.lower, q.split
Route	Memory/app/memory/retrieval/tree_retriever.py::score_node
Tags	Retrieval, Tagging
Logic	Coordinates downstream calls such as q.split.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

retrieve

Function `retrieve` executes domain-specific processing. loads and retrieves required data

Inputs: self, query, top_k. Outputs: List. Called by: no internal callers detected. Calls: scored.sort, walk.

Inputs	self, query, top_k
Outputs	List
Called by	-
Calls	walk, scored.sort
Route	Memory/app/memory/retrieval/tree_retriever.py::retrieve
Tags	Retrieval
Logic	loads and retrieves required data
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

walk

Function `walk` executes domain-specific processing. Coordinates downstream calls such as

self._score_node, walk. Inputs: node. Outputs: unspecified return value. Called by: retrieve, walk. Calls: self._score_node, walk.

Inputs	node
Outputs	unspecified return value
Called by	retrieve, walk
Calls	scored.append, self._score_node, walk
Route	Memory/app/memory/retrieval/tree_retriever.py::walk
Tags	Retrieval
Logic	Coordinates downstream calls such as self._score_node, walk.
In Workflow	Yes
Workflow Reason	Reachable from entry points and actively participates in call flow.

__get_router

Function `\_\_get\_router` loads required project data. Lazy singleton loader for the BERT

classification pipeline. Inputs: no explicit parameters. Outputs: Pipeline. Called by: predict_route, warmup. Calls: pipeline.

Inputs	-
Outputs	Pipeline

Called by	predict_route, warmup
Calls	pipeline
Route	Memory/app/memory/routing/bert_router.py::get_router
Tags	Utilities
Logic	Lazy singleton loader for the BERT classification pipeline.
In Workflow	Yes
Workflow Reason	Reachable from entry points and actively participates in call flow.

predict_route

Function `predict\_route` serves application-facing request handling. Predict intent scores using BERT semantic classification. Inputs: query, behavior_state. Outputs: Dict. Called by: no internal callers detected. Calls: _get_router, round, router.

Inputs	query, behavior_state
Outputs	Dict
Called by	-
Calls	_get_router, behavior_state.items, router, pred.lower, round, scores.items
Route	Memory/app/memory/routing/bert_router.py::predict_route
Tags	Workflow
Logic	Predict intent scores using BERT semantic classification.
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

is_model_available

Function `is\_model\_available` executes domain-specific processing. Check if the model files exist at the expected path. Inputs: no explicit parameters. Outputs: bool. Called by: no internal callers detected. Calls: MODEL_PATH.exists.

Inputs	-
Outputs	bool
Called by	-
Calls	MODEL_PATH.exists
Route	Memory/app/memory/routing/bert_router.py::is_model_available

Tags	Utilities
Logic	Check if the model files exist at the expected path.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

warmup

Function `warmup` executes domain-specific processing. Pre-load the model into memory (call during startup if desired). Inputs: no explicit parameters. Outputs: unspecified return value.

Called by: no internal callers detected. Calls: `_get_router`.

Inputs	-
Outputs	unspecified return value
Called by	-
Calls	<code>_get_router</code>
Route	Memory/app/memory/routing/bert_router.py::warmup
Tags	Utilities
Logic	Pre-load the model into memory (call during startup if desired).
In Workflow	Yes
Workflow Reason	Reachable from entry points and dispatches work to downstream functions.

apply_fallback

Function `apply\_fallback` executes domain-specific processing. Returns a system prompt modification or action directive based on the fallback. Inputs: `self`, `decision`, `retrieved_memories`.

Outputs: `str`. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	<code>self</code> , <code>decision</code> , <code>retrieved_memories</code>
Outputs	<code>str</code>
Called by	-
Calls	-
Route	Memory/app/memory/routing/fallback_manager.py::apply_fallback
Tags	Utilities
Logic	Returns a system prompt modification or action directive based on the fallback.
In Workflow	No

Workflow Reason	No entry-point-reachable call path detected for this function.
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__init__

Function `\_\_init\_\_` executes domain-specific processing. Performs `init` behavior for this module. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	-
Route	Memory/app/memory/routing/intent_classifier.py::init
Tags	Utilities
Logic	Performs `init` behavior for this module.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

score_all

Function `score\_all` executes domain-specific processing. Evaluate every intent category against *query* and return Inputs: self, query. Outputs: Dict. Called by: no internal callers detected. Calls: self._score_chitchat, self._score_factual, self._score_greeting.

Inputs	self, query
Outputs	Dict
Called by	-
Calls	query.strip, self._score_greeting, self._score_chitchat, self._score_summary, self._score_historical, self._score_relational, self._score_factual, self._score_oos
Route	Memory/app/memory/routing/intent_classifier.py::score_all
Tags	Utilities
Logic	Evaluate every intent category against *query* and return
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

classify

Function `classify` executes domain-specific processing. Score all intents, then return: Inputs: self, query. Outputs: Tuple. Called by: no internal callers detected. Calls: self._detect_conflicts, self._extract_secondary, self._normalize.

Inputs	self, query
Outputs	Tuple
Called by	-
Calls	self.score_all, self._select_dominant, self._extract_secondary, self._normalize, self._detect_conflicts
Route	Memory/app/memory/routing/intent_classifier.py::classify
Tags	Tagging
Logic	Score all intents, then return:
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_extract_secondary

Function `\_extract\_secondary` extracts and structures data. Return non-dominant intents above the secondary threshold. Inputs: self, scores, dominant. Outputs: List. Called by: no internal callers detected. Calls: IntentType, candidates.sort, self._priority_index.

Inputs	self, scores, dominant
Outputs	List
Called by	-
Calls	IntentType, scores.items, candidates.sort, self._priority_index
Route	Memory/app/memory/routing/intent_classifier.py::extract_secondary
Tags	Parsing
Logic	Return non-dominant intents above the secondary threshold.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_normalize

Function `\_normalize` filters and normalizes intermediate data. Cleans intermediate values before downstream processing. Inputs: self, scores. Outputs: Dict. Called by: no internal callers detected. Calls: round.

Inputs	self, scores
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Outputs	Dict
Called by	-
Calls	scores.values, scores.keys, round, scores.items
Route	Memory/app/memory/routing/intent_classifier.py::normalize
Tags	Utilities
Logic	Cleans intermediate values before downstream processing.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_detect_conflicts

Function `\_detect\_conflicts` executes domain-specific processing. Coordinates downstream calls such as `IntentType`, `self._priority_index`. Inputs: `self`, `scores`. Outputs: `List`. Called by: no internal callers detected. Calls: `IntentType`, `self._priority_index`.

Inputs	<code>self</code> , <code>scores</code>
Outputs	<code>List</code>
Called by	-
Calls	<code>IntentType</code> , <code>scores.items</code> , <code>self._priority_index</code>
Route	Memory/app/memory/routing/intent_classifier.py::detect_conflicts
Tags	Utilities
Logic	Coordinates downstream calls such as <code>IntentType</code> , <code>self._priority_index</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_priority_index

Function `\_priority\_index` executes domain-specific processing. Supports document retrieval and indexing flow. Inputs: `self`, `intent`. Outputs: `int`. Called by: no internal callers detected. Calls: `self._PRIORITY.index`.

Inputs	<code>self</code> , <code>intent</code>
Outputs	<code>int</code>
Called by	-
Calls	<code>self._PRIORITY.index</code>
Route	Memory/app/memory/routing/intent_classifier.py::priority_index

Tags	Retrieval
Logic	Supports document retrieval and indexing flow.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_greeting_token_count

Function `\_greeting\_token\_count` executes domain-specific processing. How many leading tokens are greeting words. Inputs: self, q. Outputs: int. Called by: no internal callers detected. Calls: self._GREETING_TIME_OF_DAY.match, self._GREETING_WORDS.match.

Inputs	self, q
Outputs	int
Called by	-
Calls	self._GREETING_TIME_OF_DAY.match, self._GREETING_WORDS.match
Route	Memory/app/memory/routing/intent_classifier.py::greeting_token_count
Tags	Utilities
Logic	How many leading tokens are greeting words.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_strip_greeting_prefix

Function `\_strip\_greeting\_prefix` executes domain-specific processing. Remove leading greeting words and return the rest. Inputs: self, q. Outputs: str. Called by: no internal callers detected. Calls: re.sub.

Inputs	self, q
Outputs	str
Called by	-
Calls	re.sub.strip, re.sub
Route	Memory/app/memory/routing/intent_classifier.py::strip_greeting_prefix
Tags	Utilities
Logic	Remove leading greeting words and return the rest.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_is_stem_chitchat

Function `\_is\_stem\_chitchat` executes domain-specific processing. Return true when a social stem has only a social tail. Inputs: self, q. Outputs: bool. Called by: no internal callers detected. Calls: match.end, q.split, self._CHITCHAT_STEM.match.

Inputs	self, q
Outputs	bool
Called by	-
Calls	self._CHITCHAT_STEM.match, q.split, q.strip, match.end, self._SOCIAL_STEM_TAIL.match
Route	Memory/app/memory/routing/intent_classifier.py::is_stem_chitchat
Tags	Utilities
Logic	Return true when a social stem has only a social tail.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_score_greeting

Function `\_score\_greeting` executes domain-specific processing. Greeting gets a high score ONLY when the query is purely social. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: q.split, self._CHITCHAT_PREFIX.match, self._CHITCHAT_STANDALONE.match.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._greeting_token_count, q.split, self._strip_greeting_prefix, self._CHITCHAT_PREFIX.match, self._CHITCHAT_STANDALONE.match
Route	Memory/app/memory/routing/intent_classifier.py::score_greeting
Tags	Utilities
Logic	Greeting gets a high score ONLY when the query is purely social.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_score_chitchat

Function `\_score\_chitchat` executes domain-specific processing. Score chitchat signals in the query. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: q.split, self._CHITCHAT_PREFIX.match, self._CHITCHAT_STANDALONE.match.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._CHITCHAT_STANDALONE.match, self._CHITCHAT_PREFIX.match, q.split, self._is_stem_chitchat, self._CONVERSATIONAL_FILLER.match, self._strip_greeting_prefix, tail.split
Route	Memory/app/memory/routing/intent_classifier.py::score_chitchat
Tags	Utilities
Logic	Score chitchat signals in the query.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_score_summary

Function `\_score\_summary` executes domain-specific processing. Score summary intent based on keyword matches. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: self._SUMMARY_PATTERNS.findall, self._strip_greeting_prefix.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._strip_greeting_prefix, self._SUMMARY_PATTERNS.findall
Route	Memory/app/memory/routing/intent_classifier.py::score_summary
Tags	Utilities
Logic	Score summary intent based on keyword matches.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_score_historical

Function `\_score\_historical` executes domain-specific processing. Score historical/episodic intent based on temporal keywords. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: self._HISTORICAL_PATTERNS.findall, self._strip_greeting_prefix.

Inputs	self, q
Outputs	float
Called by	-

Calls	self._strip_greeting_prefix, self._HISTORICAL_PATTERNS.findall
Route	Memory/app/memory/routing/intent_classifier.py::score_historical
Tags	Utilities
Logic	Score historical/episodic intent based on temporal keywords.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__score_relational

Function `\_\_score\_relational` executes domain-specific processing. Score relational/graph intent based on relationship keywords. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: self._RELATIONAL_PATTERNS.findall, self._strip_greeting_prefix.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._strip_greeting_prefix, self._RELATIONAL_PATTERNS.findall
Route	Memory/app/memory/routing/intent_classifier.py::score_relational
Tags	Utilities
Logic	Score relational/graph intent based on relationship keywords.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__score_oos

Function `\_\_score\_oos` executes domain-specific processing. Score out-of-scope intent by checking across all OOS thematic groups. Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: pat.search, self._strip_greeting_prefix.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._strip_greeting_prefix, pat.search
Route	Memory/app/memory/routing/intent_classifier.py::score_oos
Tags	Utilities
Logic	Score out-of-scope intent by checking across all OOS thematic groups.

In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_score_factual

Function `\_score\_factual` executes domain-specific processing. Score factual intent. Only triggers for queries with enough Inputs: self, q. Outputs: float. Called by: no internal callers detected. Calls: pat.search, self._CHITCHAT_PREFIX.match, self._CHITCHAT_STANDALONE.match.

Inputs	self, q
Outputs	float
Called by	-
Calls	self._strip_greeting_prefix, tail.split, self._SUMMARY_PATTERNS.search, self._HISTORICAL_PATTERNS.search, self._RELATIONAL_PATTERNS.search, pat.search, self._CHITCHAT_STANDALONE.match, self._CHITCHAT_PREFIX.match, self._is_stem_chitchat, self._CONVERSATIONAL_FILLER.match
Route	Memory/app/memory/routing/intent_classifier.py::score_factual
Tags	Utilities
Logic	Score factual intent. Only triggers for queries with enough
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_select_dominant

Function `\_select\_dominant` executes domain-specific processing. Pick the intent with the highest score. Inputs: self, scores. Outputs: Tuple. Called by: no internal callers detected. Calls: IntentType.

Inputs	self, scores
Outputs	Tuple
Called by	-
Calls	scores.values, scores.items, IntentType
Route	Memory/app/memory/routing/intent_classifier.py::select_dominant
Tags	Utilities
Logic	Pick the intent with the highest score.
In Workflow	No

Workflow Reason	No entry-point-reachable call path detected for this function.
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__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as IntentClassifier. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: IntentClassifier.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	IntentClassifier
Route	Memory/app/memory/routing/oos_detector.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as IntentClassifier.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

is_oos

Function `is\_oos` executes domain-specific processing. Returns True if the OOS score for the query exceeds the threshold. Inputs: self, query. Outputs: bool. Called by: no internal callers detected. Calls: self._classifier._score_oos.

Inputs	self, query
Outputs	bool
Called by	-
Calls	self._classifier._score_oos, query.strip
Route	Memory/app/memory/routing/oos_detector.py::is_oos
Tags	Utilities
Logic	Returns True if the OOS score for the query exceeds the threshold.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as IntentClassifier, RoutingPolicies. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: IntentClassifier, RoutingPolicies.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	IntentClassifier, RoutingPolicies
Route	Memory/app/memory/routing/query_router.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as IntentClassifier, RoutingPolicies.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

route

Function `route` serves application-facing request handling. Implements API/service handler behavior and returns normalized response data. Inputs: self, query. Outputs: RouteDecision. Called by: no internal callers detected. Calls: IntentType, RouteDecision, analyze_emotional_signals.

Inputs	self, query
Outputs	RouteDecision
Called by	-
Calls	self.intent_classifier.classify, analyze_emotional_signals, predict_route, self.intent_classifier._strip_greeting_prefix, query.strip, stripped.split, self.intent_classifier._CHITCHAT_PREFIX.match, self.intent_classifier._CHITCHAT_STANDALONE.match, self.intent_classifier._CONVERSATIONAL_FILLER.match, round, query.split, heuristic_scores.keys, bert_scores.keys, final_scores.items, suppressed_intents.append, active_intents.append, self._build_orchestration_plan, priority_order.index, IntentType, self.policies.get_policy, RouteDecision, self._build_reasoning, self._build_intent_reasoning, heuristic_scores.items, bert_scores.items
Route	Memory/app/memory/routing/query_router.py::route
Tags	Workflow

Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

_build_reasoning

Function `\_build\_reasoning` builds and composes output artifacts. Build a human-readable explanation of why this intent won. Inputs: intent, confidence, scores, secondary. Outputs: str. Called by: no internal callers detected. Calls: round.

Inputs	intent, confidence, scores, secondary
Outputs	str
Called by	-
Calls	round, scores.items, nonzero.items
Route	Memory/app/memory/routing/query_router.py::build_reasoning
Tags	Utilities
Logic	Build a human-readable explanation of why this intent won.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_build_intent_reasoning

Function `\_build\_intent\_reasoning` builds and composes output artifacts. Structured intent reasoning for debug output. Inputs: primary, secondary, scores. Outputs: str. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	primary, secondary, scores
Outputs	str
Called by	-
Calls	lines.append
Route	Memory/app/memory/routing/query_router.py::build_intent_reasoning
Tags	Utilities
Logic	Structured intent reasoning for debug output.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_build_orchestration_plan

Function `\_build\_orchestration\_plan` builds and composes output artifacts. Build an orchestration plan based on the set of active intents. Inputs: active_intents. Outputs: list. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	active_intents
Outputs	list
Called by	-
Calls	-
Route	Memory/app/memory/routing/query_router.py::build_orchestration_plan
Tags	Workflow
Logic	Build an orchestration plan based on the set of active intents.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

get_policy

Function `get\_policy` loads required project data. Retrieves module data required by downstream functions. Inputs: self, intent. Outputs: Dict. Called by: no internal callers detected. Calls: no internal downstream calls.

Inputs	self, intent
Outputs	Dict
Called by	-
Calls	-
Route	Memory/app/memory/routing/routing_policies.py::get_policy
Tags	Utilities
Logic	Retrieves module data required by downstream functions.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as os.getenv. Inputs: self, api_key, model. Outputs: unspecified return value. Called by: no internal callers detected. Calls: os.getenv.

Inputs	self, api_key, model
Outputs	unspecified return value
Called by	-

Calls	os.getenv
Route	Memory/app/core/providers/gemini_provider.py::init
Tags	Utilities
Logic	Coordinates downstream calls such as os.getenv.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_client_init

Function `\_client\_init` executes domain-specific processing. Coordinates downstream calls such as RuntimeError, genai.Client. Inputs: self. Outputs: unspecified return value. Called by: no internal callers detected. Calls: RuntimeError, genai.Client.

Inputs	self
Outputs	unspecified return value
Called by	-
Calls	RuntimeError, genai.Client
Route	Memory/app/core/providers/gemini_provider.py::client_init
Tags	Utilities
Logic	Coordinates downstream calls such as RuntimeError, genai.Client.
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

chat

Function `chat` executes domain-specific processing. Implements API/service handler behavior and returns normalized response data. Inputs: self, system, user, temperature. Outputs: str. Called by: no internal callers detected. Calls: client.models.generate_content, getattr, self._client_init.

Inputs	self, system, user, temperature
Outputs	str
Called by	-
Calls	self._client_init, client.models.generate_content, getattr
Route	Memory/app/core/providers/gemini_provider.py::chat
Tags	Workflow

Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

__init__

Function `\_\_init\_\_` executes domain-specific processing. Coordinates downstream calls such as `os.getenv`. Inputs: `self`, `api_key`, `model`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `os.getenv`.

Inputs	<code>self</code> , <code>api_key</code> , <code>model</code>
Outputs	unspecified return value
Called by	-
Calls	<code>os.getenv</code>
Route	<code>Memory/app/core/providers/openai_provider.py::init</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>os.getenv</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

_client_init

Function `\_client\_init` executes domain-specific processing. Coordinates downstream calls such as `OpenAI`, `RuntimeError`. Inputs: `self`. Outputs: unspecified return value. Called by: no internal callers detected. Calls: `OpenAI`, `RuntimeError`.

Inputs	<code>self</code>
Outputs	unspecified return value
Called by	-
Calls	<code>RuntimeError</code> , <code>OpenAI</code>
Route	<code>Memory/app/core/providers/openai_provider.py::client_init</code>
Tags	Utilities
Logic	Coordinates downstream calls such as <code>OpenAI</code> , <code>RuntimeError</code> .
In Workflow	No
Workflow Reason	No entry-point-reachable call path detected for this function.

chat

Function `chat` executes domain-specific processing. Implements API/service handler behavior and returns normalized response data. Inputs: self, system, user, temperature. Outputs: str. Called by: no internal callers detected. Calls: client.chat.completions.create, self._client_init.

Inputs	self, system, user, temperature
Outputs	str
Called by	-
Calls	self._client_init, client.chat.completions.create
Route	Memory/app/core/providers/openai_provider.py::chat
Tags	Workflow
Logic	Implements API/service handler behavior and returns normalized response data.
In Workflow	No
Workflow Reason	Supporting helper in a module that is reachable from workflow entry points.

Chapter 3 — Execution Workflow

1. `run_once` (`app/jobs/sleep_consolidation_job.py`) — orchestrates workflow execution steps
2. `run_forever` (`app/jobs/sleep_consolidation_job.py`) — orchestrates workflow execution steps
3. `run` (`app/workflow/orchestrator.py`) — orchestrates workflow execution steps
4. `search` (`app/storage/vector_store.py`) — Handles search stage transition
5. `cosine` (`app/storage/vector_store.py`) — Handles cosine stage transition
6. `analyze` (`app/memory/orchestration/attention_engine.py`) — Handles analyze stage transition
7. `to_dict` (`app/memory/retrieval/context_tree.py`) — Handles to dict stage transition
8. `walk` (`app/memory/retrieval/context_tree.py`) — Handles walk stage transition
9. `load` (`app/memory/retrieval/persistent_tree_store.py`) — loads and retrieves required data
10. `rebuild` (`app/memory/retrieval/persistent_tree_store.py`) — builds and composes output artifacts
11. `retrieve` (`app/memory/retrieval/tree_retriever.py`) — loads and retrieves required data
12. `walk` (`app/memory/retrieval/tree_retriever.py`) — Handles walk stage transition
13. `predict_route` (`app/memory/routing/bert_router.py`) — Predict intent scores using BERT semantic classification
14. `warmup` (`app/memory/routing/bert_router.py`) — Pre-load the model into memory (call during startup if desired)
15. `chat` (`app/api/chat.py`) — Handles chat stage transition
16. `recent` (`app/api/debug.py`) — Handles recent stage transition
17. `stats` (`app/api/debug.py`) — Handles stats stage transition
18. `interactions` (`app/api/debug.py`) — Handles interactions stage transition
19. `graph_data` (`app/api/graph.py`) — JSON graph data: center node + neighbors + labeled edges
20. `graph_view` (`app/api/graph.py`) — Interactive visual graph viewer using vis-network
21. `debug_route` (`app/api/router_api.py`) — Test the intent classifier and router logic without running the full pipeline
22. `router_stats` (`app/api/router_api.py`) — Returns general stats or config of the router
23. `get_tree` (`app/api/tree.py`) — Retrieves required data in `get_tree`
24. `mcp_search` (`app/mcp/server.py`) — Handles mcp search stage transition

The workflow starts from `run_once`

Chapter 4 — Dependency Map

Dependency relation distribution:

Relation	Count
calls	601
imports	247
defines	214
inherits_from	21

From	Relation	To	Source File
chat	calls	run	app/api/chat.py
interactions	calls	list_interactions	app/api/debug.py
recent	calls	list_recent	app/api/debug.py
stats	calls	interaction_stats	app/api/debug.py
stats	calls	is_available	app/api/debug.py
stats	calls	list_recent	app/api/debug.py
graph_data	calls	rich_neighborhood	app/api/graph.py
debug_route	calls	route	app/api/router_api.py
get_tree	calls	to_dict	app/api/tree.py
get_tree	calls	load	app/api/tree.py
chat	calls	_client_init	app/core/providers/gemini_provider.py
chat	calls	_client_init	app/core/providers/openai_provider.py
run_once	calls	compact	app/jobs/sleep_consolidation_job.py
run_once	calls	decay	app/jobs/sleep_consolidation_job.py
run_once	calls	reflect	app/jobs/sleep_consolidation_job.py
run_once	calls	save	app/jobs/sleep_consolidation_job.py
run_once	calls	build_from_memories	app/jobs/sleep_consolidation_job.py
run_once	calls	store	app/jobs/sleep_consolidation_job.py
run_once	calls	prune	app/jobs/sleep_consolidation_job.py
run_once	calls	list_recent	app/jobs/sleep_consolidation_job.py
run_once	calls	prune_interactions	app/jobs/sleep_consolidation_job.py
run_once	calls	update_status	app/jobs/sleep_consolidation_job.py

From	Relation	To	Source File
startup_event	calls	start_background_jobs	app/main.py
startup_event	calls	initialize	app/main.py
mcp_search	calls	retrieve	app/mcp/server.py
mcp_recent	calls	list_recent	app/mcp/server.py
analyze_emotional_signals	calls	search	app/memory/behavior/emotional_analyzer.py
compact	calls	_similarity	app/memory/consolidation/compaction_engine.py
compact	calls	add	app/memory/consolidation/compaction_engine.py
_has_negation_flip	calls	_tokenize	app/memory/extraction/contradiction_detector.py
_jaccard	calls	_tokenize	app/memory/extraction/contradiction_detector.py
_split_atomic_facts	calls	add	app/memory/extraction/memory_extractor.py
validate	calls	_similarity	app/memory/extraction/memory_validator.py
_similarity	calls	_tokenize	app/memory/extraction/memory_validator.py
validate	calls	find_contradictions	app/memory/extraction/memory_validator.py
analyze	calls	analyze_emotional_signals	app/memory/orchestration/attention_engine.py
_score_importance	calls	search	app/memory/orchestration/interaction_filter.py
retrieve	calls	load	app/memory/retrieval/hybrid_retriever.py
retrieve	calls	rank	app/memory/retrieval/hybrid_retriever.py
retrieve	calls	search	app/memory/retrieval/hybrid_retriever.py
retrieve	calls	neighborhood	app/memory/retrieval/hybrid_retriever.py
retrieve	calls	get_memory	app/memory/retrieval/hybrid_retriever.py
retrieve	calls	list_recent	app/memory/retrieval/hybrid_retriever.py
save	calls	to_dict	app/memory/retrieval/persistent_tree_store.py
rank	calls	add	app/memory/retrieval/retrieval_ranker.py
build_from_memories	calls	add_path	app/memory/retrieval/tree_indexer.py
<module>	calls	resolve	app/memory/routing/bert_router.py

From	Relation	To	Source File
_score_factual	calls	search	app/memory/routing/intent_classifier.py
_score_oos	calls	search	app/memory/routing/intent_classifier.py
is_oos	calls	_score_oos	app/memory/routing/oos_detector.py
route	calls	analyze_emotional_signals	app/memory/routing/query_router.py
route	calls	predict_route	app/memory/routing/query_router.py
route	calls	_strip_greeting_prefix	app/memory/routing/query_router.py
route	calls	classify	app/memory/routing/query_router.py
route	calls	get_policy	app/memory/routing/query_router.py
retrieve	calls	search	app/memory/supermemory.py
store	calls	add	app/memory/supermemory.py
store	calls	add_memory	app/memory/supermemory.py
store	calls	delete_memory	app/memory/supermemory.py
add	calls	embed	app/storage/faiss_vector_store.py
search	calls	embed	app/storage/faiss_vector_store.py
_load	calls	load	app/storage/faiss_vector_store.py
add	calls	initialize	app/storage/faiss_vector_store.py
search	calls	initialize	app/storage/faiss_vector_store.py
rich_neighborhood	calls	add	app/storage/graph_store.py
add_relation	calls	add_edge	app/storage/graph_store.py
neighborhood	calls	get_edges	app/storage/graph_store.py
prune	calls	prune_edges	app/storage/graph_store.py
rich_neighborhood	calls	get_memory	app/storage/graph_store.py
_get_connection	calls	initialize	app/storage/sqlite_store.py
search	calls	embed	app/storage/vector_store.py

From	Relation	To	Source File
upsert	calls	embed	app/storage/vector_store.py
run	calls	chat	app/workflow/orchestrator.py
run	calls	reinforce	app/workflow/orchestrator.py
run	calls	extract	app/workflow/orchestrator.py
run	calls	validate	app/workflow/orchestrator.py
run	calls	analyze	app/workflow/orchestrator.py
_log_interaction	calls	should_store	app/workflow/orchestrator.py
run	calls	build_context	app/workflow/orchestrator.py
run	calls	save	app/workflow/orchestrator.py
run	calls	build_from_memories	app/workflow/orchestrator.py
run	calls	apply_fallback	app/workflow/orchestrator.py
run	calls	route	app/workflow/orchestrator.py
run	calls	get_policy	app/workflow/orchestrator.py
run	calls	retrieve	app/workflow/orchestrator.py
run	calls	store	app/workflow/orchestrator.py
_build_graph_edges	calls	add_relation	app/workflow/orchestrator.py
_build_graph_edges	calls	connect_related_memories	app/workflow/orchestrator.py
_log_interaction	calls	add_interaction	app/workflow/orchestrator.py
_log_interaction	calls	list_recent_interactions	app/workflow/orchestrator.py
run	calls	list_recent	app/workflow/orchestrator.py
run	calls	update_status	app/workflow/orchestrator.py

Architecture snapshot (module interaction summary):

From Module	To Module	Interaction	Links
app/api/debug.py	app/storage/sqlite_store.py	calls	5

From Module	To Module	Interaction	Links
app/storage/graph_store.py	app/storage/sqlite_store.py	calls	4
app/workflow/orchestrator.py	app/storage/sqlite_store.py	calls	4
app/jobs/sleep_consolidation_job.py	app/storage/sqlite_store.py	calls	3
app/memory/supermemory.py	app/storage/sqlite_store.py	calls	3
app/memory/retrieval/hybrid_retriever.py	app/storage/sqlite_store.py	calls	3
app/memory/supermemory.py	app/storage/faiss_vector_store.py	calls	2
app/storage/faiss_vector_store.py	app/storage/sqlite_store.py	calls	2
app/storage/faiss_vector_store.py	app/core/embeddings.py	calls	2
app/storage/vector_store.py	app/core/embeddings.py	calls	2
app/workflow/orchestrator.py	app/memory/supermemory.py	calls	2
app/workflow/orchestrator.py	app/storage/graph_store.py	calls	2
app/memory/extraction/contradiction_detector.py	app/memory/extraction/memory_validator.py	calls	2
app/memory/extraction/memory_validator.py	app/memory/extraction/contradiction_detector.py	calls	2
app/memory/routing/intent_classifier.py	app/storage/faiss_vector_store.py	calls	2
app/memory/routing/intent_classifier.py	app/storage/sqlite_store.py	calls	2
app/memory/routing/intent_classifier.py	app/storage/vector_store.py	calls	2
app/memory/routing/query_router.py	app/memory/routing/intent_classifier.py	calls	2
app/main.py	app/storage/faiss_vector_store.py	calls	1
app/main.py	app/storage/sqlite_store.py	calls	1

Chapter 5 — Testing Scenarios

Function-level sample scenarios:

Each scenario is presented as a structured test card (Input / Expected / Negative / Trace).

startup_event	
Input	no arguments
Expected	Returns successful stage output with `startup event` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

root	
Input	no arguments
Expected	Returns successful stage output with `root` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

chat	
Input	payload='value'
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

recent	
Input	no arguments
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

stats	
Input	no arguments
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body

Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

interactions

Input	limit=10
Expected	Returns successful stage output with `interactions` logic applied and result fields populated.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

Chapter 5 — Testing Scenarios (continued)

Function-level sample scenarios:

graph_data	
Input	memory_id='value'
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

graph_view	
Input	memory_id='value'
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

debug_route	
Input	req='value'
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

router_stats	
Input	no arguments
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

get_tree	
Input	no arguments
Expected	Returns successful stage output with `get tree` logic applied and result fields populated.
Negative	missing required request payload fields or malformed query/body

Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

embed

Input	self='value', text='sample_text', dim='value'
Expected	Returns List with `embed` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

Chapter 5 — Testing Scenarios (continued)

Function-level sample scenarios:

__init__	
Input	self='value'
Expected	Initializes required dependencies and object state without runtime errors.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

chat	
Input	self='value', system='value', user='value', temperature='value'
Expected	Returns API-ready str payload with stable keys and serializable values.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

__init__	
Input	self='value'
Expected	Initializes required dependencies and object state without runtime errors.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

run_once	
Input	self='value'
Expected	Returns successful stage output after orchestrating downstream stage calls.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

run_forever	
Input	self='value', interval_seconds='value'
Expected	Returns successful stage output after orchestrating downstream stage calls.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

start_background_jobs	
Input	no arguments
Expected	Returns successful stage output with `start background jobs` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

Chapter 5 — Testing Scenarios (continued)

Function-level sample scenarios:

mcp_search	
Input	query='cache eviction strategy'
Expected	Returns successful stage output with `mcp search` logic applied and result fields populated.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

mcp_recent	
Input	no arguments
Expected	Returns API-ready successful stage output payload with stable keys and serializable values.
Negative	missing required request payload fields or malformed query/body
Negative Expected	Returns validation-focused error response with actionable message.
Trace Hint	Trace excerpt: request validation/runtime error in API handler.

_passes_admission	
Input	self='value', m='value'
Expected	Returns bool with `passes admission` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

store	
Input	self='value', memories='value'
Expected	Returns List with `store` logic applied and result fields populated.
Negative	invalid input type/shape
Negative Expected	Raises validation error or returns deterministic fallback output.
Trace Hint	Trace excerpt: TypeError or ValueError in core function logic.

retrieve	
Input	self='value', query='cache eviction strategy', limit=10
Expected	Returns List with required records loaded and normalized.
Negative	unavailable document id or empty retrieval result
Negative Expected	Returns controlled fallback/empty result and logs retrieval failure.
Trace Hint	Trace excerpt: RuntimeError or ValueError from retrieval call path.

__init__	
Input	self='value', dim='value', index_path='./state/example.json'
Expected	Initializes required dependencies and object state without runtime errors.
Negative	database/index unavailable or invalid persistence payload
Negative Expected	Returns controlled storage error and preserves existing state.
Trace Hint	Trace excerpt: sqlite/faiss/io runtime error with operation context.

Workflow-level sample scenarios:

Happy path workflow	
Input	Project folder path + valid configuration
Expected	Workflow executes sequence: run_once -> run_forever -> run -> search -> cosine
Negative	Project path with missing state files or failed upstream API calls
Negative Expected	Pipeline reports stage failure with partial artifacts and actionable error context.
Trace Hint	Trace excerpt: RuntimeError in run_stage with failing stage name and root exception.

Robustness path	
Input	Project folder path + partial metadata / missing docs
Expected	Workflow still completes with fallback descriptions and generated report.
Negative	Corrupt intermediate JSON + missing required retrieval outputs
Negative Expected	Workflow skips/flags invalid segments and emits validation diagnostics.
Trace Hint	Trace excerpt: ValueError/json.JSONDecodeError during stage handoff parsing.

Chapter 6 — Summary

Total modules found: 54

Total classes found: 62

Total functions found: 160

Entry points identified: 20

Most connected nodes:

- app/memory/routing/query_router.py::route: 37 connections
- app/workflow/orchestrator.py::run: 28 connections
- app/storage/sqlite_store.py::SQLiteStore: 27 connections
- app/memory/behavior/emotional_analyzer.py::analyze_emotional_signals: 27 connections
- app/memory/consolidation/compaction_engine.py::compact: 22 connections
- app/workflow/orchestrator.py: 19 connections
- app/memory/routing/intent_classifier.py::IntentClassifier: 19 connections
- app/memory/retrieval/hybrid_retriever.py::retrieve: 18 connections
- app/jobs/sleep_consolidation_job.py::run_once: 17 connections
- app/memory/extraction/memory_extractor.py::extract: 17 connections